S3 (SIMPLE STORAGE SERVICE)

Amazon S3 (Simple Storage Service) is an object storage service provided by Amazon Web Services. It is used to store and retrieve any amount of data from anywhere over the internet. Amazon S3 offers high scalability, durability, and availability for data storage. It is commonly used for backups, file storage, and application data.

- Amazon S3 is an independent storage service provided by Amazon Web Services.
- It is a global service that can be accessed from anywhere over the internet.
- Amazon S3 is an object storage service used to store data as objects in buckets.
- Users can create, upload, store, and retrieve data at any time.
- Amazon S3 is commonly used for file storage, backups, and application data.

S3 BUCKET

- An S3 Bucket is a container used to store data in Amazon S3.
- It is used to store unstructured data such as images, videos, documents, and backups.
- Data in an S3 bucket is stored in the form of objects.
- Each object consists of the data, metadata, and a unique key.
- Every bucket has a globally unique name in AWS.
- A bucket is created in a specific AWS region but can be accessed globally.
- Buckets help to organize, manage, and secure data in Amazon S3.

TYPES OF S3 BUCKETS BASED ON ACCESS

1. Private Bucket

- A Private Bucket is accessible only to authorized users.
- By default, all S3 buckets are private.
- Access is controlled using IAM policies and bucket policies.
- Data in private buckets is secure and protected.
- Commonly used for backups, logs, and confidential data.

Types of Data Stored in a Private S3 Bucket

(Data not supposed to be shared on the internet)

- Sensitive business data (reports, documents, backups)
- Personal user information (IDs, records, credentials)

- Application configuration files
- Database backups and logs
- Internal project files
- Source code and confidential media files

2. Public Bucket

- A Public Bucket allows anyone on the internet to access objects.
- Public access must be explicitly enabled.
- Used for static website hosting and public files.
- Requires careful configuration to avoid security risks.
- Not recommended for sensitive data.

Types of Data Stored in a Public S3 Bucket

(Data directly exposed on the internet)

- Static website files (HTML, CSS, JavaScript)
- Web pages for public access
- Images, videos, and media files
- Downloadable files (PDFs, software installers)
- Public datasets and open resources

TWO LEVELS OF PERMISSIONS IN AMAZON S3

1. Bucket-Level Permission

Bucket-level permissions control who can access the bucket and what actions they can perform on it.

- Applied to the entire bucket
- Controls actions like list, upload, or delete objects
- Managed using bucket policies and IAM policies

2. Object-Level Permission

Object-level permissions control access to individual objects (files) inside a bucket.

- Applied to specific objects
- Allows fine-grained access control
- Managed using ACLs and object policies

ACL (ACCESS CONTROL LIST) IN AMAZON S3

An ACL (Access Control List) is a permission mechanism used in Amazon Web Services S3 to control who can access buckets and objects and what actions they can perform, such as read or write.

ACL at Bucket Level (Enable / Disable)

At the bucket level, ACL determines whether public or cross-account access is allowed.

Explanation:

- When ACL public access is enabled, the bucket can be made public by granting permissions.
- When ACL public access is disabled, the bucket remains private.
- By default, ACL public access is disabled to improve security.

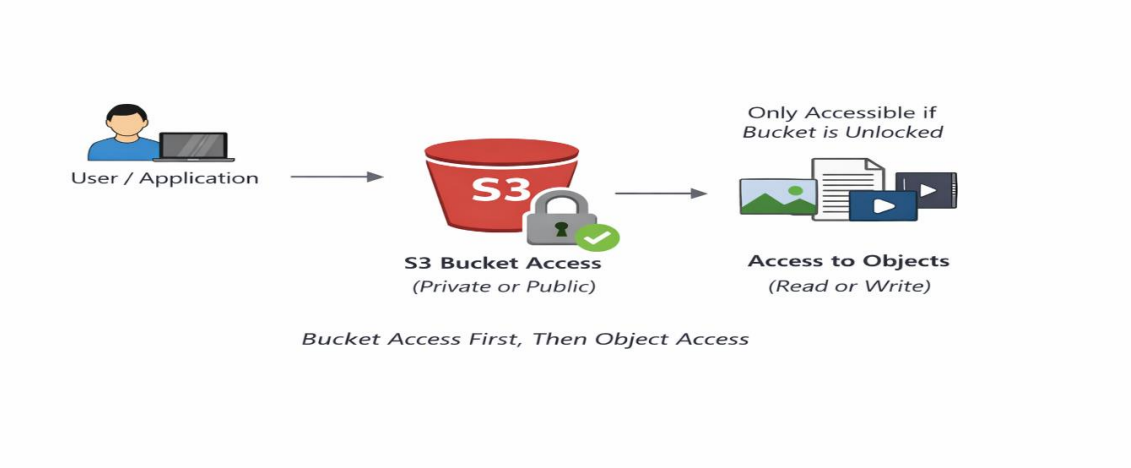
Use of ACL in a Bucket level

- Grants read or write permissions to specific AWS accounts
- Controls access to the entire bucket
- Helps in cross-account access scenarios
- Less commonly used compared to IAM and bucket policies

Use of ACL at Object Level

- Controls access to individual objects (files)
- Allows sharing specific objects with other AWS accounts
- Used when fine-grained object access is required
- Common in legacy or cross-account object sharing

S3 Access Flow: Bucket Permission Before Object Access



- A user or application requests access to data stored in Amazon S3.
- S3 first checks bucket-level permissions.
- If bucket access is denied, access to all objects inside the bucket is blocked.
- If bucket access is allowed, S3 then checks object-level permissions.
- Object permissions (ACL or policy) decide read or write access.
- Only when both bucket and object permissions allow, the object becomes accessible.

VERSIONING IN S3

Versioning is a feature in Amazon S3 that keeps multiple versions of the same object in a bucket whenever the object is uploaded, updated, or deleted.

When versioning is enabled:

- Every time you upload a file with the same name, S3 saves it as a new version
- Older versions are not overwritten or lost
- You can restore or download any previous version

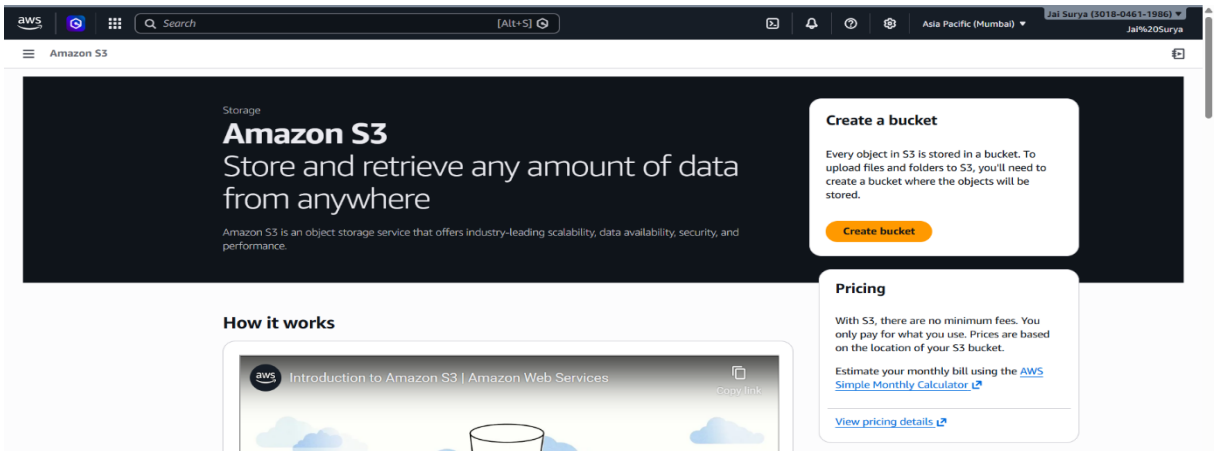
When versioning is suspended :

- Existing object versions are kept and not deleted
- New uploads with the same file name do not create new versions
- New files overwrite the latest version
- Older versions can still be viewed and restored
- No new Version ID is generated for new uploads

HANDS ON SCREENSHOTS

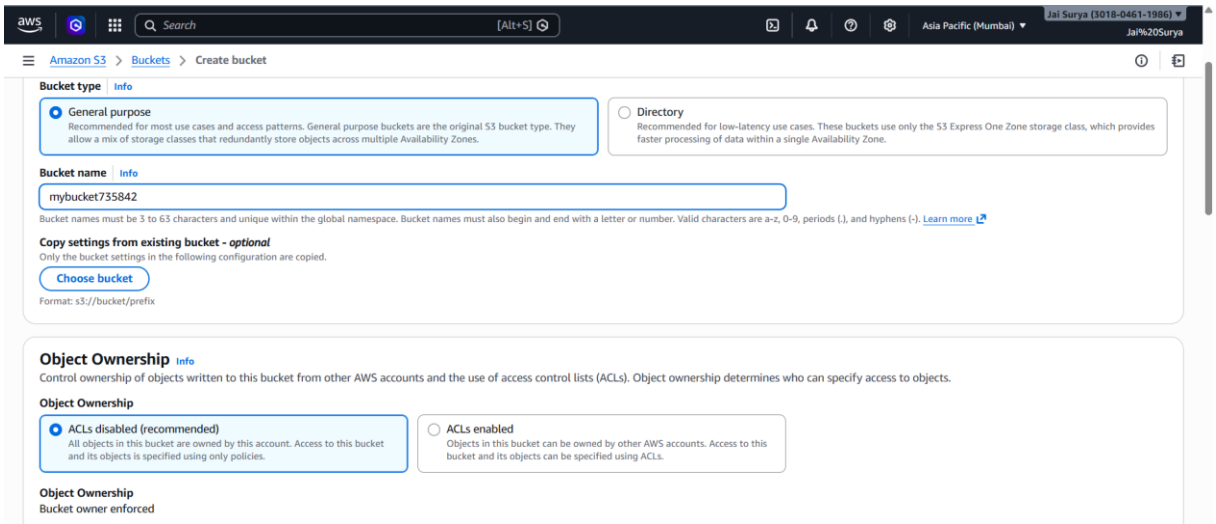
Screenshot 1: Create Bucket Page

1. Open the AWS Management Console
2. Navigate to Amazon S3
3. Click Create bucket
4. The Create bucket page is displayed



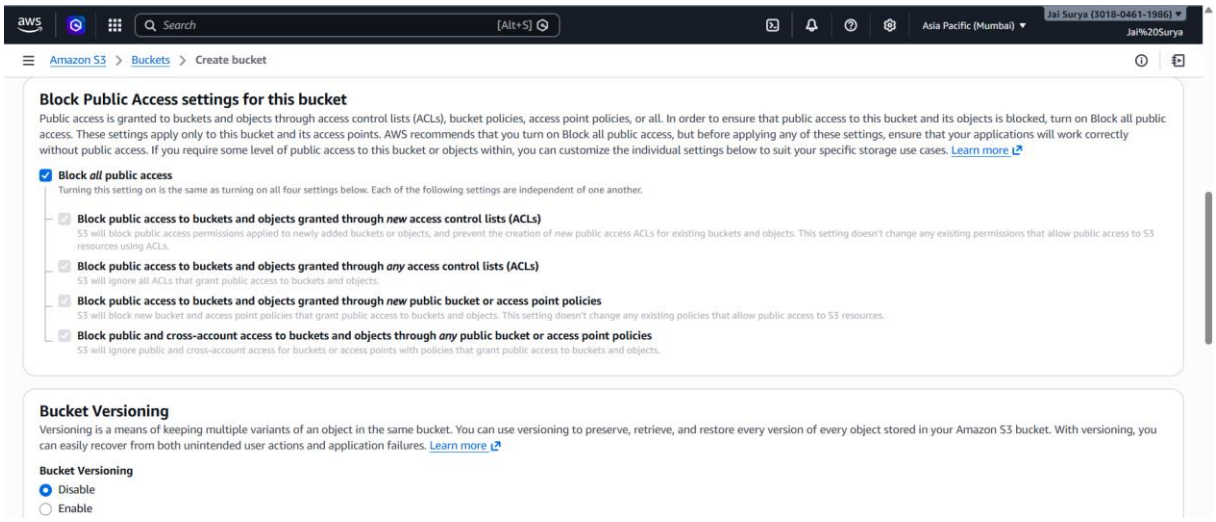
Screenshot 2: Bucket Type, Name, and Object Ownership

1. Select Bucket type as General purpose
2. Enter a unique bucket name
3. Choose the required AWS Region
4. Set Object Ownership to ACLs disabled (Bucket owner enforced)
5. This ensures the bucket owner has full control over all objects



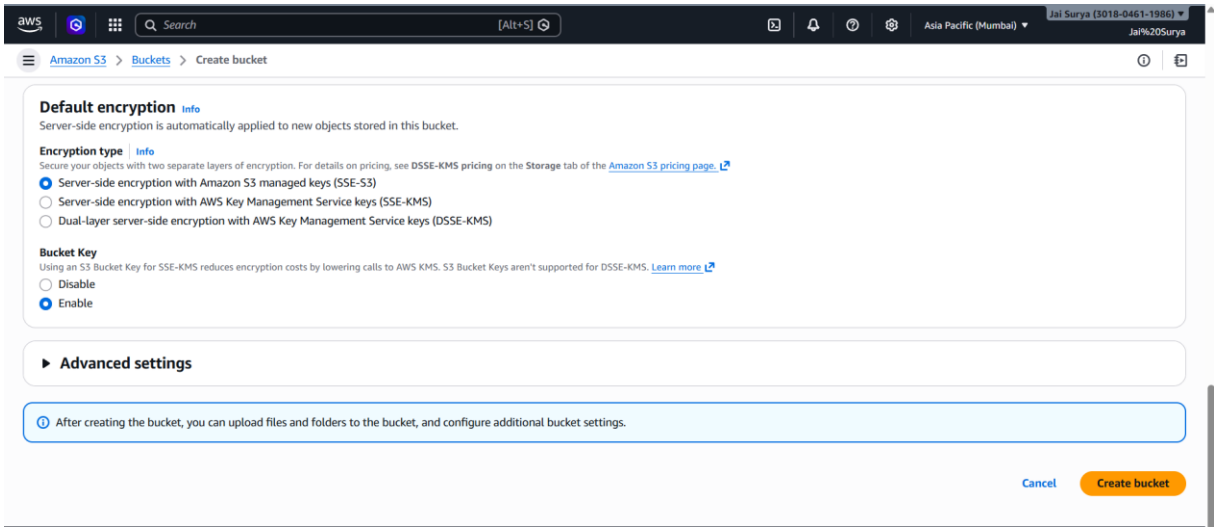
Screenshot 3: Block Public Access and Versioning Settings

1. Enable Block all public access
2. This makes the bucket private
3. Keep Bucket Versioning set to Disabled
4. Review the security settings



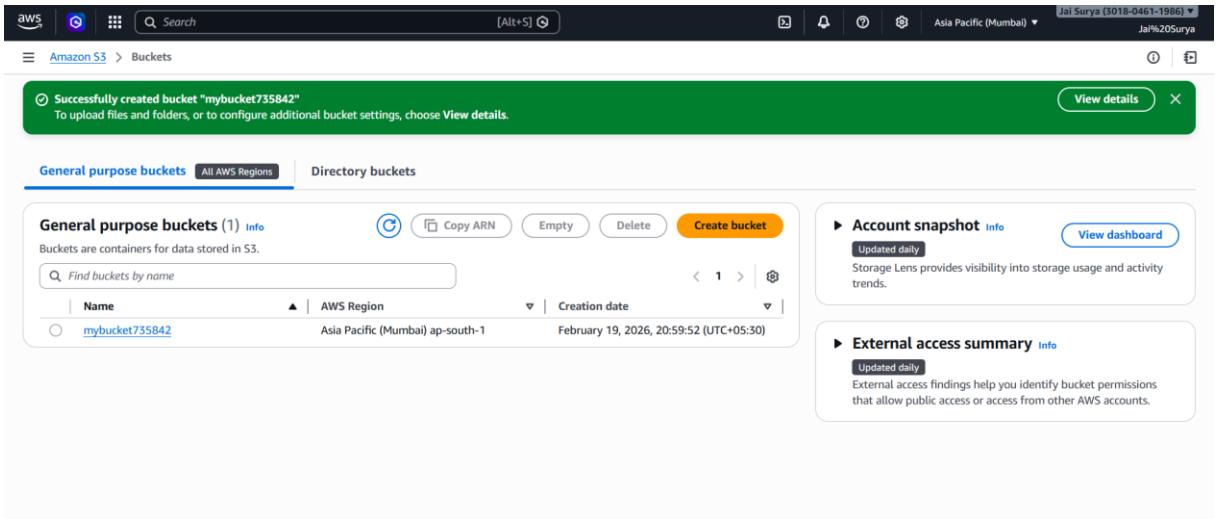
Screenshot 4: Encryption Settings

1. Select Server-side encryption
2. Choose encryption type as SSE-S3
3. Keep Bucket Key set to Disabled
4. This enables AWS-managed encryption for stored objects



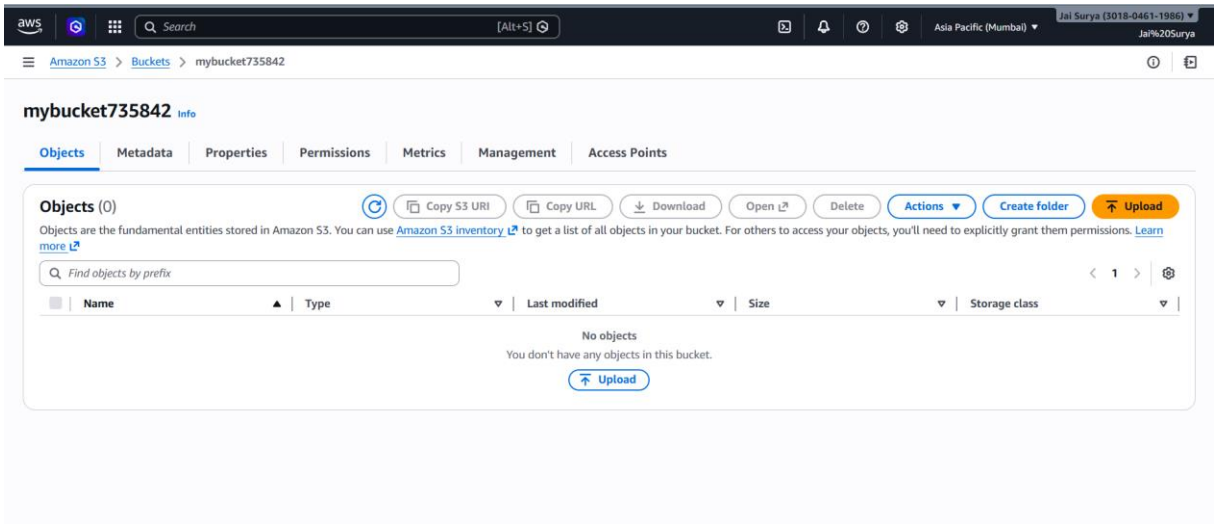
Screenshot 5: Bucket Created Successfully

1. Review all bucket configurations
2. Click **Create bucket**
3. The S3 bucket is created successfully
4. Bucket appears in the S3 bucket list



Screenshot 6: Upload Object to the Bucket

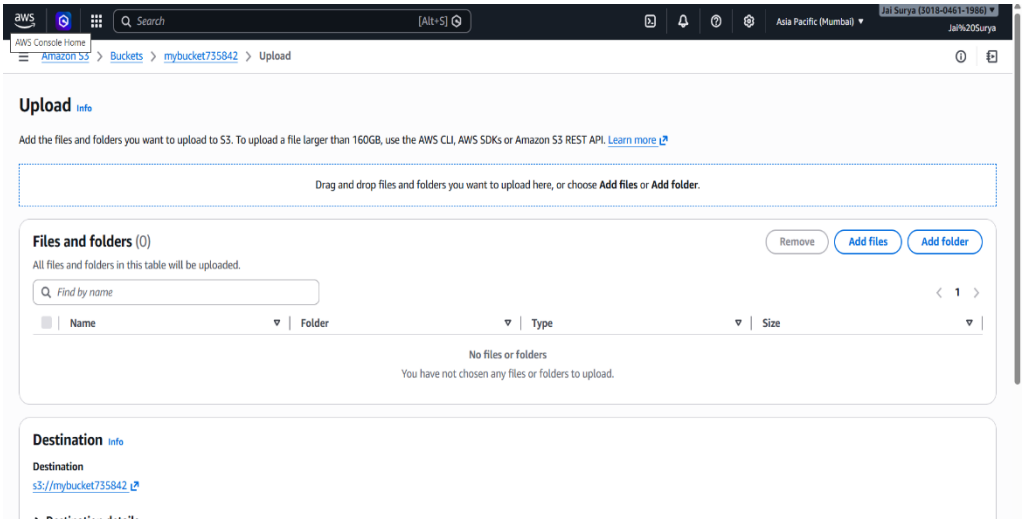
1. Open the created S3 bucket
2. Click Upload



Screenshot 7: Choose File for Upload

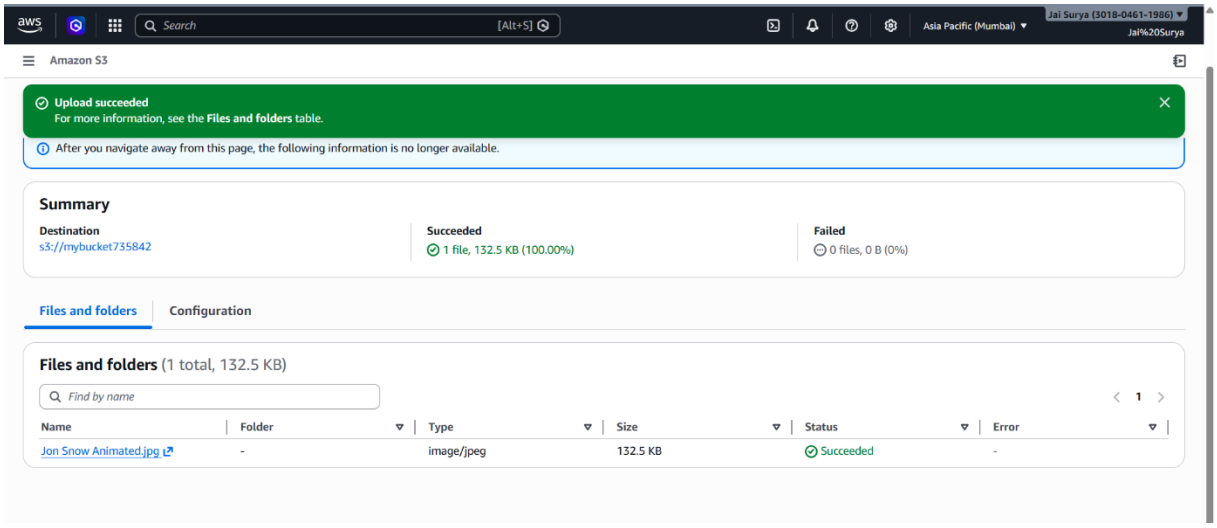
1. Click Add files
2. Select the required file from the local system

3. Confirm the selected file
4. Proceed with the upload process



Screenshot 8: Object Uploaded Successfully

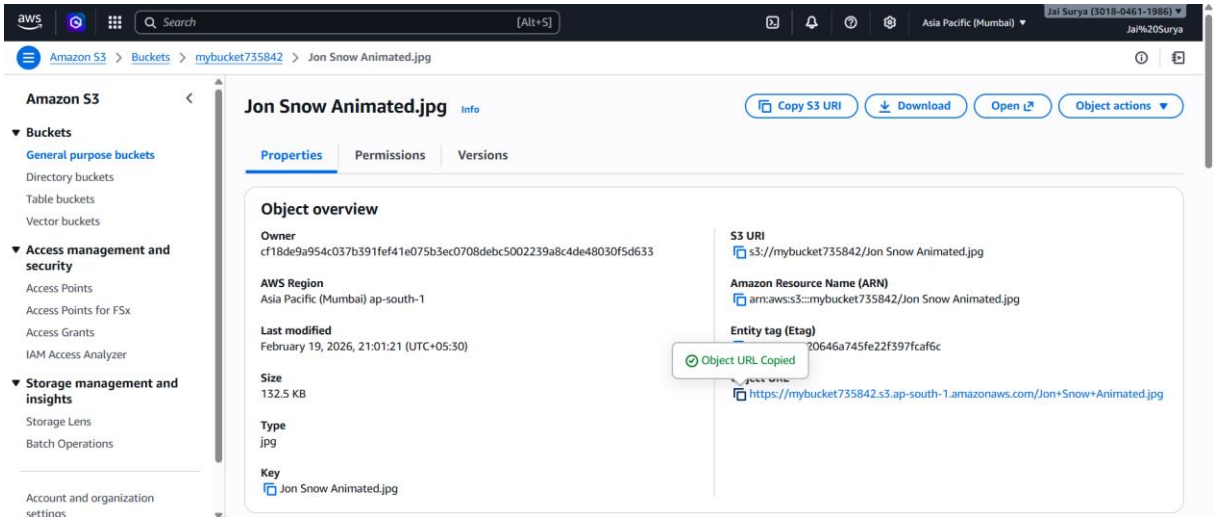
1. File upload process completed
2. Upload status shows Succeeded
3. Object is now stored in the S3 bucket
4. File is visible in the bucket object list



Screenshot 9: Copy Object URL

1. Select the uploaded object
2. Click Copy URL

3. Object URL is copied to the clipboard



Screenshot 10: Access Object URL in Browser (Private Bucket)

1. Open a new browser tab
2. Paste the copied object URL
3. Press Enter to access the file
4. Access fails because the bucket is currently private

This XML file does not appear to have any style information associated with it. The document tree is shown below.

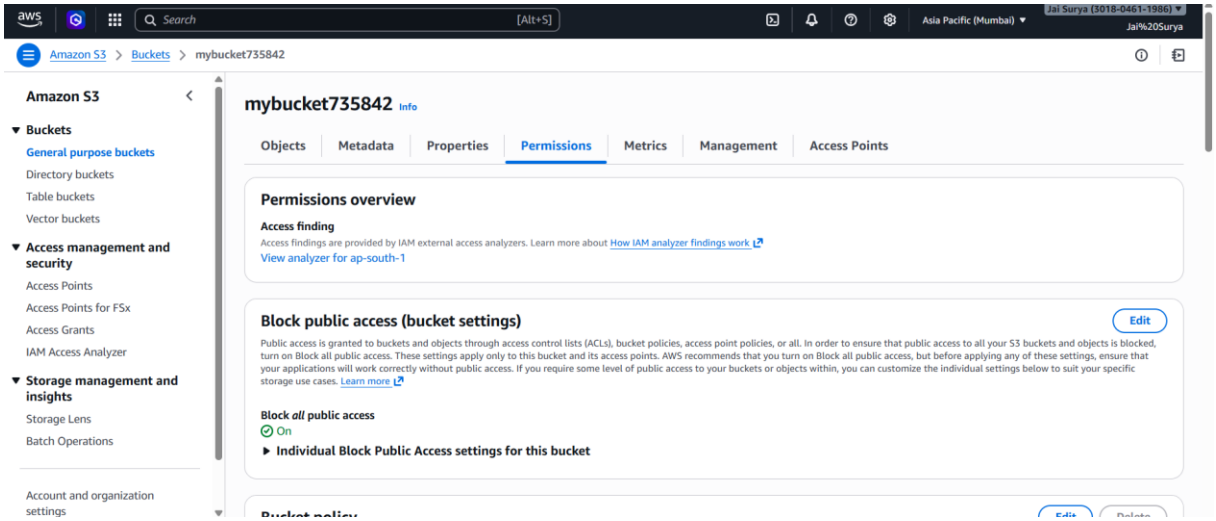

```
▼<Error>
  <Code>AccessDenied</Code>
  <Message>Access Denied</Message>
  <RequestId>445HR1PVVKC8QPP2</RequestId>
  <HostId>4h1SMffU5dfSeW/74P7WeD6laL7o6FpC8mgLG8X8eqvak2Bmi1cCOIsYer8oh2wXTSCtFpMP1vONBFTunkBa8J4Sy/a7sU7s</HostId>
</Error>
```

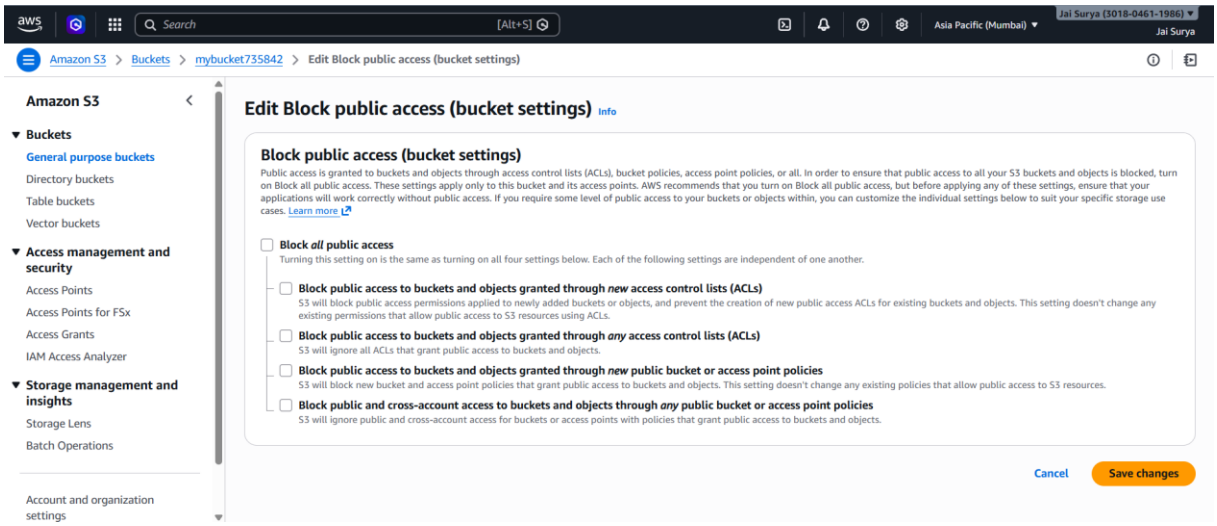

CHANGE BUCKET FROM PRIVATE TO PUBLIC

Screenshot 11& 12 : Change Bucket from Private to Public

1. Open the S3 bucket and go to the Permissions tab
2. Click Edit under Block public access
3. Uncheck “Block all public access”

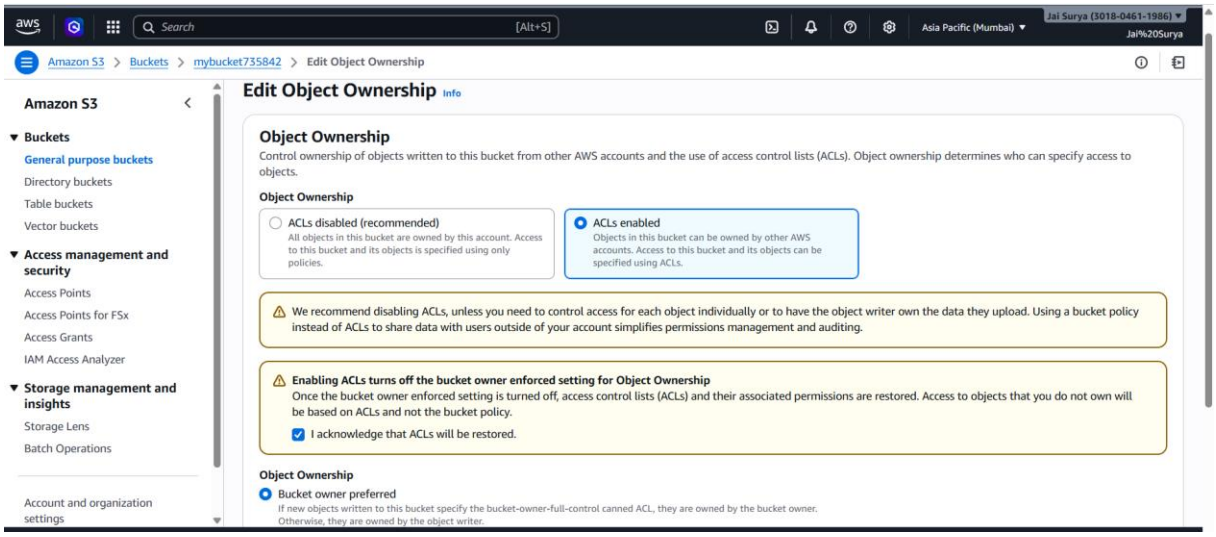
4. Confirm the warning and save changes
5. Bucket is now publicly accessible





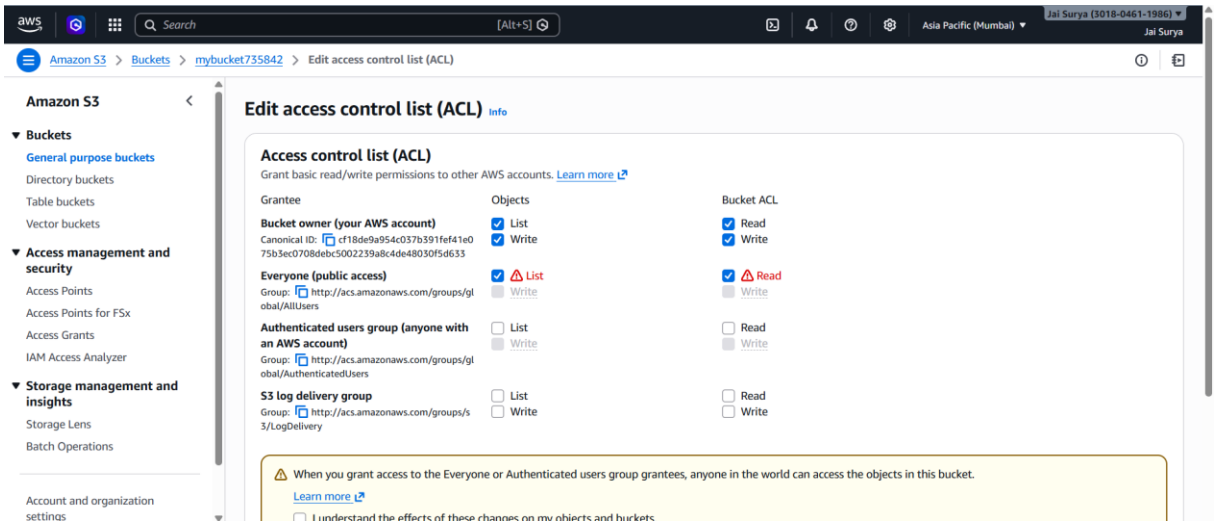
Screenshot 13: Enable ACL

1. Open the S3 bucket and go to Permissions → Access Control List (ACL)
2. Enable ACL for the bucket or object
3. Grant read access to Everyone (public) if needed
4. Click Save changes
5. This allows the object to be accessed publicly via its URL



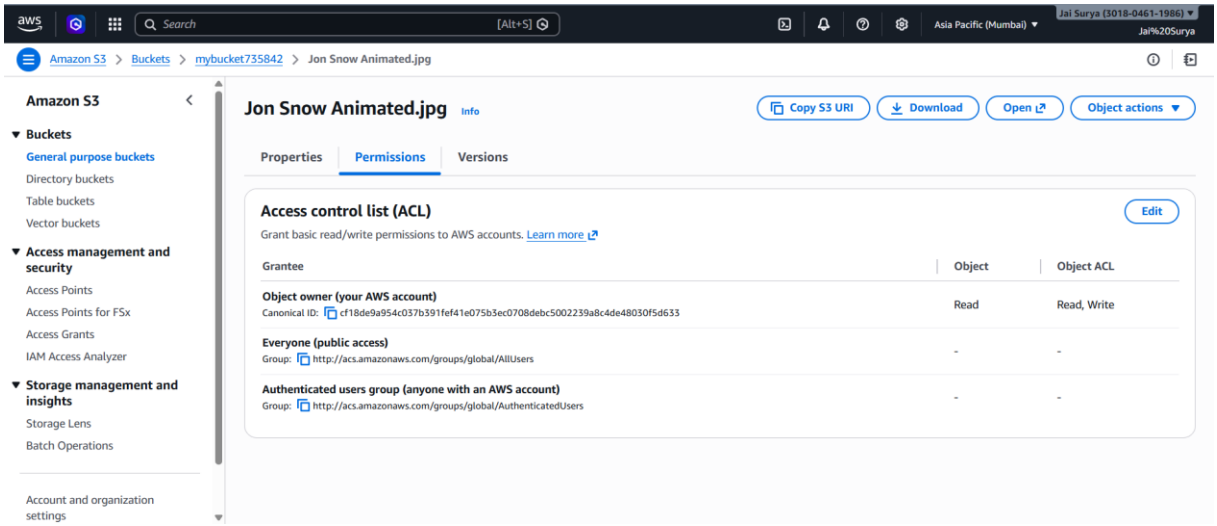
Screenshot 14: Edit Bucket ACL

1. Open the S3 bucket
2. Go to Permissions → Access Control List (ACL)
3. Click Edit under Bucket ACL
4. Enable Read access for Everyone (public access) if you want the bucket to be public
5. Click Save changes



Screenshot 15: Object Permissions

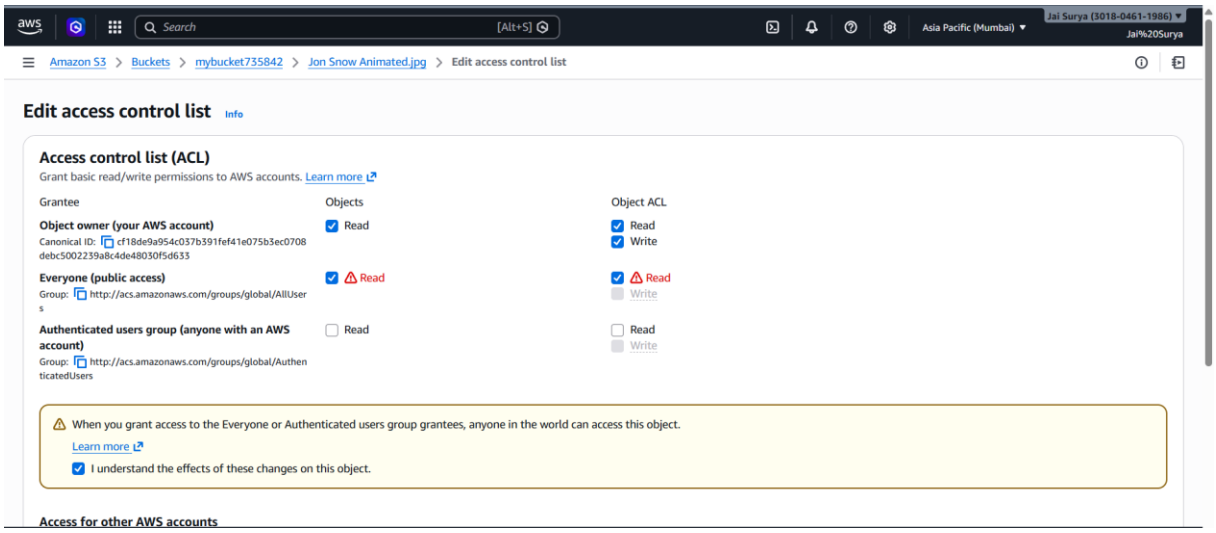
1. Open the S3 bucket and select the uploaded object
2. Go to Permissions



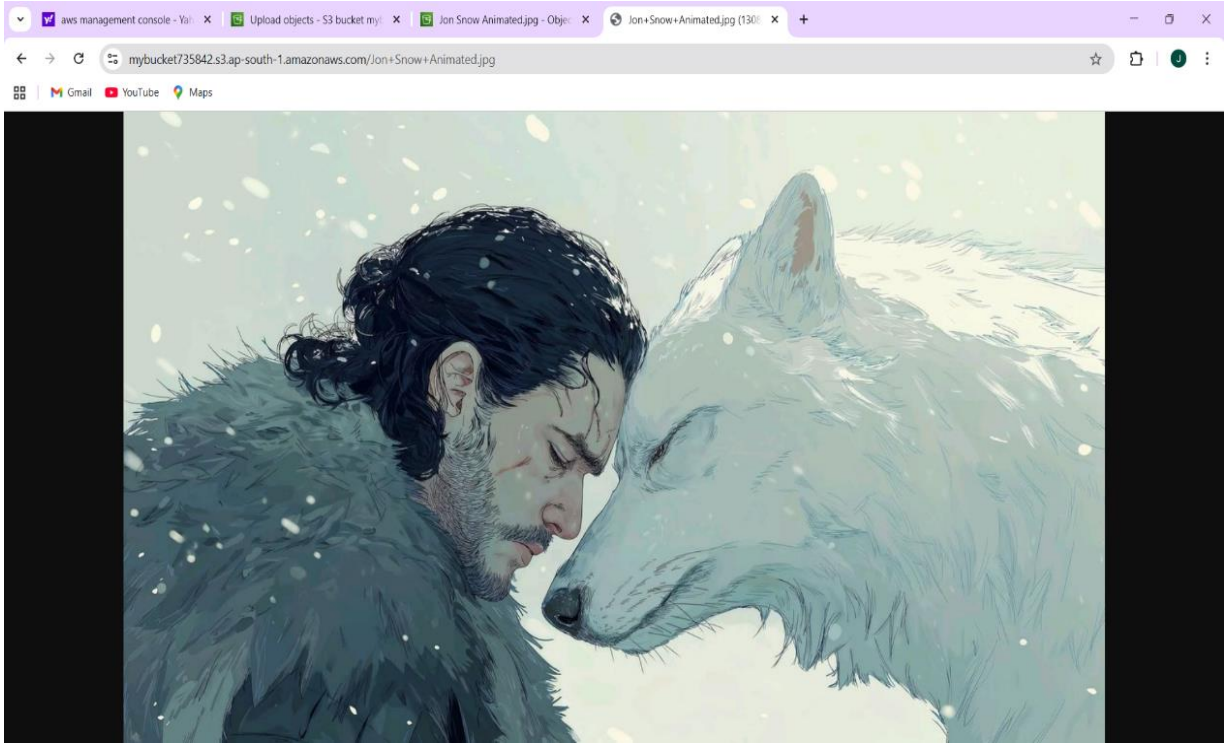
Screenshot 16: Edit ACL for Object

1. Select the uploaded object in the S3 bucket
2. Go to Permissions → Access Control List (ACL)
3. Click Edit
4. Enable Read access for Everyone (public access) if you want the object to be publicly accessible
5. Click Save changes

Result: The object can now be accessed publicly via its URL, following both bucket-level and object-level permissions.



Screenshot 17: Object Accessed Successfully in Browser

1. Open a new browser tab
2. Paste the object URL copied earlier
3. Press Enter
4. The object (file/image/page) loads successfully

Result: The object is now publicly accessible after changing the bucket to public and updating ACLs/permissions.